

Solve

CSE331 - Automata and Computability

Quiz 5

Section __ (Set A)

Date:

Full name:

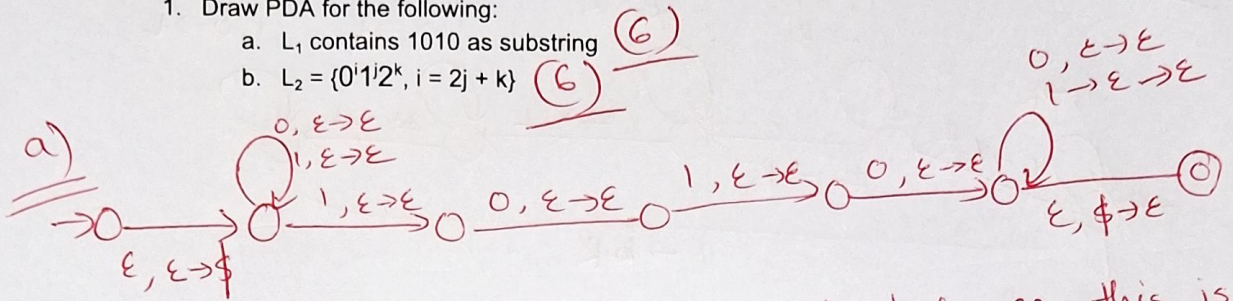
Roll:

Total Marks: 20

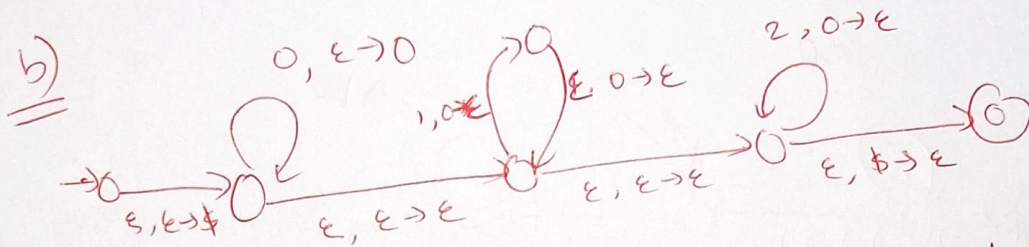
1. Draw PDA for the following:

a. L_1 contains 1010 as substring

b. $L_2 = \{0^i 1^j 2^k, i = 2j + k\}$



Draw the NFA with no change to stack, as this is a regular language.

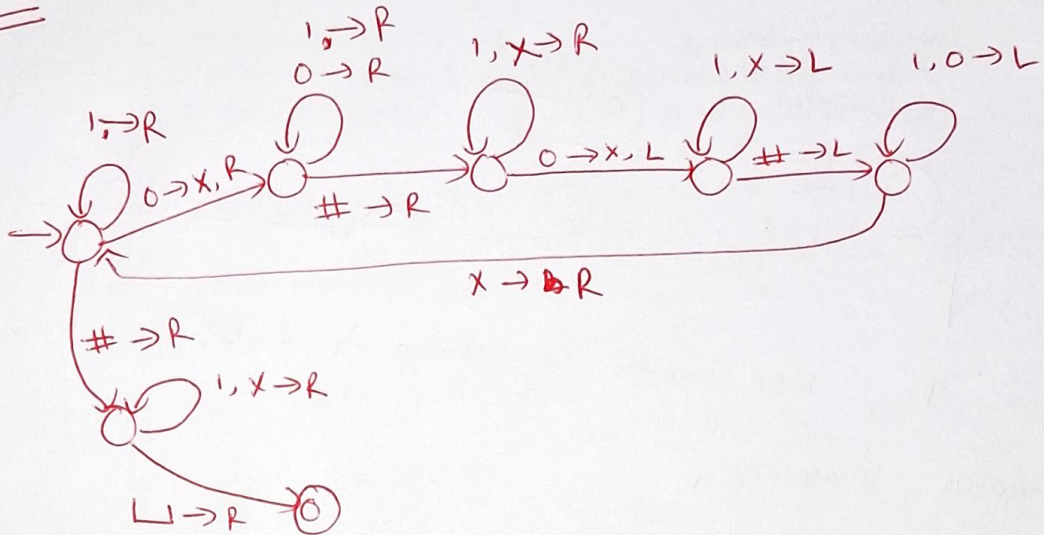


there will be two 0s for each 1 and one 0 for each 2.

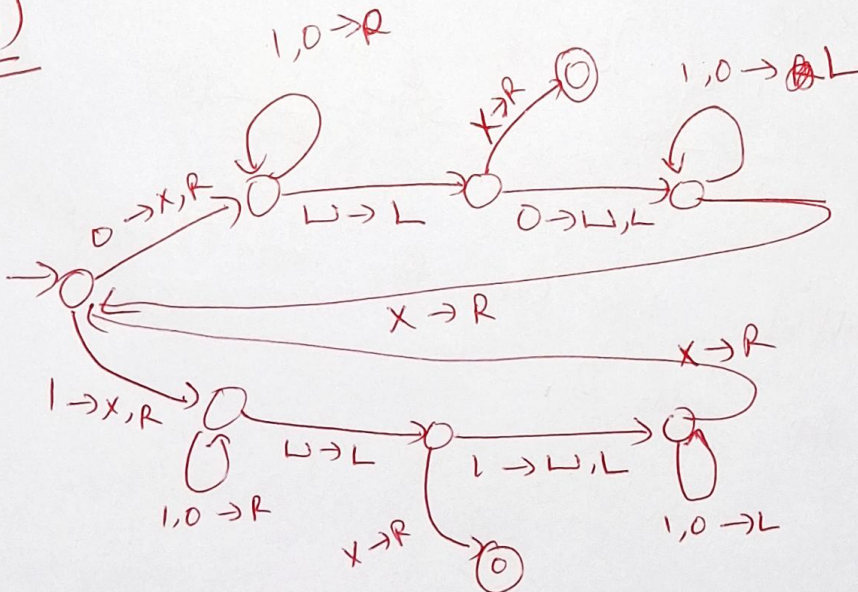
2. Draw Turing Machine for the following:

- $L_3 = \{w_1 \# w_2, \text{ number of 0's in } w_1 \text{ is same as the number of 0's in } w_2\}$
- L_4 detects odd palindromes.

a)



b)



Solve

CSE331 - Automata and Computability
Quiz 5
Section __ (Set B)

Date:

Full name:

Roll:

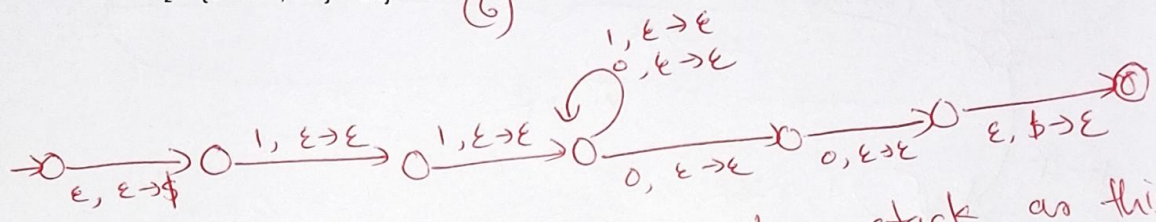
Total Marks: 20

1. Draw PDA for the following:

a. L_1 starts with 11 and ends with 00 (6)

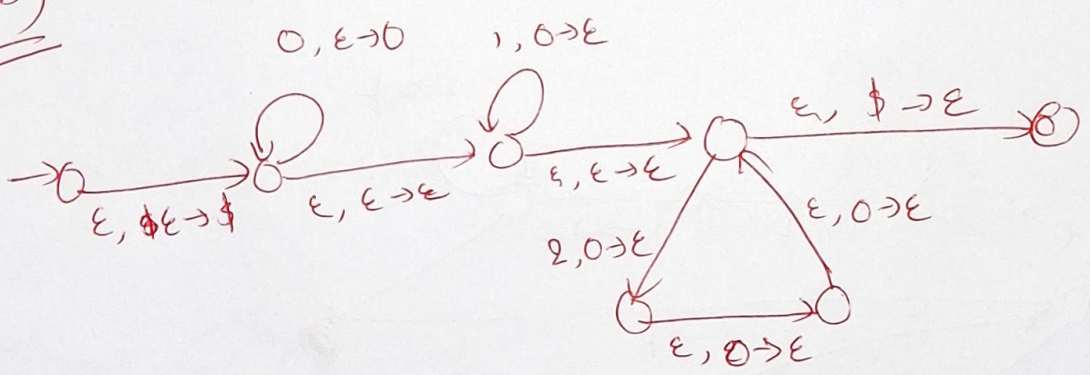
b. $L_2 = \{0^i 1^j 2^k, i = j + 3k\}$ (6)

a)



Draw the NFA with no change to stack as this is a regular language.

b)



There will be one 0 for one 1 and three 0s for one 2.

2. Draw Turing Machine for the following:

- $L_3 = \{w_1 \# w_2, \text{ number of 1's in } w_1 \text{ is same as the number of 0's in } w_2\}$
- L_4 detects even palindromes.

